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TECHNICAL NOTE

Analysis of Perishable-Inventory Systems with Censored Demand Data

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We consider a multiperiod inventory system of a perishable product with unobservable lost sales. Demand distribution parameters are unknown and are updated periodically using the Bayesian approach based on the censored historical sales data. We develop an explicit expression of the first-order condition for optimality that demonstrates the key trade-off of the problem. The result generalizes partial characterizations of this trade-off in the literature. It shows that the myopic solution is a lower bound on the optimal inventory level. It also enables us to quantify the expected marginal value of information.

Subject classifications: stochastic inventory control; censored demand data; Bayesian models; exact analysis; sample-path approach.
Area of review: Stochastic Models.
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1. Introduction

We study a periodic-review, single-item inventory system over a finite planning horizon. The item is perishable; excess inventory in each period is salvaged at a price lower than the purchasing price. Demand in excess of inventory on hand in each period is lost at a penalty. The type of demand distribution is known, but its parameters are only subject to a prior belief. The prior is updated periodically with the past sales data according to the Bayesian formula. However, because the lost demand cannot be observed or recorded, the historical sales data does not truly represent the demand and are termed censored data (censored by inventory availability). Therefore, the challenge here is how to use censored data intelligently to determine the right inventory levels so that the total expected cost in the planning horizon is minimized.

This system turns out to be very complex. To analyze its dynamics, we need to be able to trace its history. In fact, one of our contributions here is to devise adequate notation for it. This notation helps us use a sample-path analysis to derive an explicit expression of the first-order condition for determining the optimal inventory levels. The result provides a precise characterization of the key trade-off of the problem. That is, although stocking more units increases the likelihood of overstocking and therefore increases the cost of the current period, it reduces the chance of demand being censored, and hence improves the accuracy of demand estimation for future periods. The latter effect, in turn, helps make better future inventory decisions and thus reduce future costs, which is termed *the informational benefit*. From this expression, it follows immediately that the myopic inventory level, which minimizes the inventory cost in the current period but ignores the impact of learning from observed sales data on future periods, is a lower bound on the optimal inventory level. The expression also enables us to quantify the expected marginal value of information. Finally, it lays a foundation for future development of simple solution techniques to facilitate real operations. (Some progress has already been made along this line—see Lu et al. 2004—although much more work is needed.) Section 2 introduces the model and assumptions, §3 presents the main results. The online appendix contains the proofs of the main results and some lemmas used in the proofs. An electronic companion to this paper is available as part of the online version that can be found at http://or.journal.informs.org/.

Several papers have studied related dynamic programming problems. Harpaz et al. (1982) analyze a problem in the statistics setting (not in the inventory context); Lariviere and Porteus (1999) examine an inventory setting with a specific demand distribution that belongs to a family called *newsvendor distributions* (see Braden and Freimer 1991).

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Ding et al. (2002) study a general demand distribution in a two-period inventory model (see Lu et al. 2005 and Ding et al. 2005 for comments about this work). One major finding of these works is that the myopic solution is a lower bound on the optimal solution. The result was established through a partial characterization of the optimality condition, an inequality. (Recall that we are able to obtain a complete characterization, i.e., an equality.) In addition, with the specific demand distribution, Lariviere and Porteus (1999) obtain several other structural insights. For example, they show that the state-space reduction technique developed by Scarf (1959, 1960) and Azoury (1985) for the backlogging model can be extended to the case of censored data and can also hold for nonperishable products. They also show that, in the presence of censored data, stocking out could be less preferred to inventory surplus because the latter provides more accurate demand information than the former. Finally, it is worth mentioning that Nahmias (1994) and Agrawal and Smith (1996) study how to estimate demand parameters with censored data.

2. Model and Preliminaries

We consider a periodic-review inventory-control problem for a perishable product over a T-period planning horizon. The demands are stochastic and are independent and identically distributed across periods. The demand distribution has an unknown parameter vector $\theta \in \Theta$, and we apply the Bayesian approach to update our knowledge about it over periods. Let Z_t denote the demand in period t (we use the capital notation to represent a random variable and the lower case, z_t, to represent its realization). For any given θ, Z_t has a probability density function $f(z_t \mid \theta)$ and a cumulative distribution function $F(z_t \mid \theta)$. At the beginning of period t, our knowledge about θ is a prior distribution $\pi_t(\theta)$. With this prior distribution, we can compute the marginal density function m_t and the marginal cumulative distribution function M_t for Z_t as

$$m_t(z_t) = \int_\Theta f(z_t \mid \theta)\pi_t(\theta)\, d\theta,$$

$$M_t(z_t) = \int_\Theta F(z_t \mid \theta)\pi_t(\theta) d\theta.$$

The event sequence in each period is the following: At the beginning of the period, based on the latest information, we decide the order quantity. The order arrives immediately (the analysis can be extended to a constant replenishment lead time). During the period, demand occurs and is fulfilled entirely if there is enough inventory. Any inventory leftovers will be discarded with a salvage value at the end of the period. If the demand exceeds the inventory level, however, the unsatisfied portion is lost and cannot be observed, which incurs a shortage-penalty cost. Finally, we use the observed sales information to update the demand distribution for the next period, which will be detailed below.

The sales information observed in any period t consists of two parts, denoted by

$$o_t = (o_{t,1}, o_{t,2}).$$

Here, $o_{t,1}$ is the sales quantity that can take any nonnegative value; $o_{t,2}$ is the observation status taking value e or c, indicating exact or censored, respectively.

For an illustration, suppose that the inventory level (after ordering) at the beginning of period t is y and the demand realization is z_t. If $y > z_t$, then the sales quantity is z_t and the observation of the demand is exact. Otherwise, the sales quantity is y and the observation is censored. Thus, o_t is determined by y and z_t, which we express as

$$o_t = y \otimes z_t \triangleq \begin{cases} (z_t, e) & \text{if } y > z_t, \\ (y, c) & \text{if } y \leqslant z_t. \end{cases} \tag{1}$$

For expositional simplicity, in the remainder of the paper, we use the shorthand notation z_t^e and y^c to represent (z_t, e) and (y, c), respectively. For example, with the sales information 10^e, we know that the demand realized in this period is exactly 10. However, with 10^c, we can only tell that the demand is not less than 10.

At the end of period t, based on the observed sales information o_t, we update π_t to π_{t+1}, the posterior distribution of θ. We define

$$l(o_t \mid \theta) = \begin{cases} f(o_{t,1} \mid \theta) & \text{if } o_{t,2} = e, \\ 1 - F(o_{t,1} \mid \theta) & \text{if } o_{t,2} = c, \end{cases}$$

as the likelihood function of the sales information o_t for any given θ. Based on the Bayes formula, we have

$$\pi_{t+1}(\theta \mid \pi_t, o_t) = \frac{l(o_t \mid \theta)\pi_t(\theta)}{\int_\Theta l(o_t \mid \theta')\pi_t(\theta')\, d\theta'}. \tag{2}$$

Applying the above technique recursively, we can use the prior at t, π_t, and observations afterward, $\mathbf{o}_{t,t+i-1} = (o_t, \ldots, o_{t+i-1})$, to compute the prior for any future period $t+i$ as follows:

$$\pi_{t+i}(\theta \mid \pi_t, \mathbf{o}_{t,t+i-1}) = \frac{\Pi_{j=0}^{i-1} l(o_{t+j} \mid \theta)\pi_t(\theta)}{\int_\Theta \Pi_{j=0}^{i-1} l(o_{t+j} \mid \theta')\pi_t(\theta')\, d\theta'}. \tag{3}$$

See Lemma 1 in the online appendix for the proof of the above result, as well as the marginal density function $m_{t+i}(z_{t+i} \mid \pi_t, \mathbf{o}_{t,t+i-1})$ for period $t+i$.

Let c, h, and p be the unit ordering cost, the unit salvage value, and the unit shortage-penalty cost in each period, respectively (the analysis can be extended to nonstationary cost parameters). Without loss of generality, we assume that $p > c > h$. The objective is to minimize the total expected cost over the T-period planning horizon.

We now formulate the problem as a dynamic program. The state in period t is our current knowledge about θ, which is π_t. The decision is the order quantity y. Let $V_t(\pi_t)$ be the optimal expected cost from period t to T given that

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the prior distribution of θ is π_t at the beginning of period t. Then,

$$V_{T+1}(\pi_{T+1}) = 0, \tag{4}$$

$$V_t(\pi_t) = \min_{y \geqslant 0}\{G_t(y, \pi_t)\}, \quad 1 \leqslant t \leqslant T, \tag{5}$$

where

$$\begin{aligned} G_t(y, \pi_t) &= C_t(y, \pi_t) + E[V_{t+1}(\pi_{t+1})] \\ &= C_t(y, \pi_t) + \int_0^y V_{t+1}(\pi_{t+1}(\cdot \mid z_t^e))m_t(z_t)\,dz_t \\ &\quad + V_{t+1}(\pi_{t+1}(\cdot \mid y^c))[1 - M_t(y)], \end{aligned} \tag{6}$$

$$C_t(y, \pi_t) = \int_0^{+\infty} r(y, z_t)m_t(z_t)\,dz_t, \tag{7}$$

$$r(y, z_t) = cy - h(y - z_t)^+ + p(z_t - y)^+.$$

Let $S_t^*(\pi_t)$ be the optimal inventory level in period t given π_t, i.e.,

$$S_t^*(\pi_t) = \arg\min_{y \geqslant 0} G_t(y, \pi_t). \tag{8}$$

As indicated by (1)–(3), we see that the inventory decision y in period t affects the sales observation o_t, which in turn affects the demand-distribution updates from t on. In addition, as shown in (4) and (5), y affects the inventory decisions and costs in all future periods as well. To highlight the impact of the inventory decision at t on period $t+i$, we often replace π_{t+i} by $(\pi_t, \mathbf{o}_{t,\,t+i-1})$, provided the latter information is given (recall that $\mathbf{o}_{t,\,t+i-1}$ depends on both inventory decisions and demand realizations in periods t through $t+i-1$). We can do so because π_{t+i} can be obtained from this information, as shown in (3).

More specifically, we equate $C_{t+i}(y, \pi_t, \mathbf{o}_{t,\,t+i-1})$ with $C_{t+i}(y, \pi_{t+i})$, $G_{t+i}(y, \pi_t, \mathbf{o}_{t,\,t+i-1})$ with $G_{t+i}(y, \pi_{t+i})$, and $S_{t+i}^*(\pi_t, \mathbf{o}_{t,\,t+i-1})$ with $S_{t+i}^*(\pi_{t+i})$, with the understanding that π_{t+i} is computed from π_t and $\mathbf{o}_{t,\,t+i-1}$, i.e.,

$$\begin{aligned} &C_{t+i}(y, \pi_t, \mathbf{o}_{t,\,t+i-1}) \\ &\quad = \int_0^{+\infty} r(y, z_{t+i})m_{t+i}(z_{t+i} \mid \pi_t, \mathbf{o}_{t,\,t+i-1})\,dz_{t+i}, \\ &G_{t+i}(y, \pi_t, \mathbf{o}_{t,\,t+i-1}) \\ &\quad = C_{t+i}(y, \pi_t, \mathbf{o}_{t,\,t+i-1}) + E[V_{t+i+1}(\pi_{t+i+1})], \\ &S_{t+i}^*(\pi_t, \mathbf{o}_{t,\,t+i-1}) = \arg\min_{y \geqslant 0} G_{t+i}(y, \pi_t, \mathbf{o}_{t,\,t+i-1}), \end{aligned}$$

for $0 \leqslant i \leqslant T - t$. Note that when $i = 0$, the information vector (π_t, o_t, o_{t-1}) becomes vacuous, so the definition reduces to $G_t(y, \pi_t)$, $C_t(y, \pi_t)$, and $S_t^*(\pi_t)$ given in (6), (7), and (8).

To make the exposition more compact, we introduce the following notation for history of information:

$$\begin{aligned} &H_{t+1} = (\pi_t, o_t), \\ &H_{t+i} = (\pi_t, o_t, S_{t+1}^*(\widehat{H}_{t+1}) \otimes z_{t+1}, \ldots, S_{t+i-1}^*(\widehat{H}_{t+i-1}) \otimes z_{t+i-1}), \\ &\qquad\qquad 2 \leqslant i \leqslant T - t, \end{aligned}$$

where $\widehat{H}_{t+j}$ is the information used to compute the order quantity in period $t+j$.

With this notation, we define a suboptimal policy that has the following expected cost-to-go:

$$J_{t+1}(S_{t+1}^*(\widehat{H}_{t+1}), H_{t+1}) = \sum_{i=1}^{T-t} E[C_{t+i}(S_{t+i}^*(\widehat{H}_{t+i}), H_{t+i})],$$

where $\widehat{H}_{t+i} = \widehat{H}_{t+i-1} \cup (S_{t+i-1}^*(\widehat{H}_{t+i-1}) \otimes z_{t+i-1})$ for all $i \geqslant 2$. Under this policy, in *any* period from period $t+1$ on, while the actual information is indeed H_{t+i}, an "optimal" order quantity is placed as if the obtained information were $\widehat{H}_{t+i}$. However, if the optimal policy is followed from period $t+1$ on given H_{t+1}, the resulting cost-to-go is $G_{t+1}(S_{t+1}^*(H_{t+1}), H_{t+1})$ due to (8). Therefore, it is straightforward that

$$G_{t+1}(S_{t+1}^*(H_{t+1}), H_{t+1}) \leqslant J_{t+1}(S_{t+1}^*(\widehat{H}_{t+1}), H_{t+1}). \tag{9}$$

3. Main Results

First, we present an explicit expression for the derivative of $G_t(y, \pi_t)$.

THEOREM 1. *The first derivative of $G_t(y, \pi_t)$ is*

$$\begin{aligned} G_t'(y, \pi_t) &= C_t'(y, \pi_t) \\ &\quad + \big[G_{t+1}(S_{t+1}^*(H_{t+1}^e), H_{t+1}^e) \\ &\qquad - J_{t+1}(S_{t+1}^*(H_{t+1}^c), H_{t+1}^e)\big]m_t(y), \end{aligned} \tag{10}$$

where $H_{t+1}^e = (\pi_t, y^e)$ and $H_{t+1}^c = (\pi_t, y^c)$.

Note that $G_t'(y, \pi_t)$ measures the marginal cost of having additional inventory dy (where $dy > 0$) at the beginning of period t. In fact, in deriving G_t', we compare two scenarios—the inventory level after ordering in period t is y (scenario 1) or $y + dy$ (scenario 2). The two scenarios have a cost difference in period t as follows:

$$C_t(y + dy, \pi_t) - C_t(y, \pi_t) = C_t'(y, \pi_t)\,dy + o(dy),$$

which gives the first term $C_t'(y, \pi_t)$ in the right-hand side of (10), the marginal inventory cost in the current period.

We now show that the second term in the right-hand side of (10) is precisely the marginal benefit of having additional inventory dy for the case $z_t = y$. (The second term is negative, so its absolute value represents benefit.) We discuss three cases of the demand realization z_t: $z_t < y$, $y < z_t < y + dy$, and $z_t > y + dy$.

Case 1. $z_t < y$. In the two scenarios, the observations in period t are both z_t^e, which in turn gives the same prior knowledge at the beginning of period $t+1$, i.e., $\pi_{t+1} = (\pi_t, z_t^e)$. Thus, there is no cost difference in period $t+1$ and beyond between the two scenarios.

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Case 2. $y < z_t < y + dy$. In scenario 1, we obtain a censored observation of y^c. Because *any* inventory decision in future periods is based on this observation, the resulting cost is $J_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1})$. In scenario 2, we obtain an exact observation of y^e and accordingly make inventory decisions afterwards, which results in a cost of $G_{t+1}(S^*_{t+1}(H^e_{t+1}), H^e_{t+1})$. The probability of $y < z_t < y + dy$ is approximately $m_t(y)\,dy$. In short, compared with scenario 1, the additional inventory dy in scenario 2 allows us to obtain a more accurate observation in period t. This gives a cost reduction (i.e., an informational value) of $J_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1}) - G_{t+1}(S^*_{t+1}(H^e_{t+1}), H^e_{t+1})$, which leads to the second term in the right-hand side of (10).

Case 3. $z_t > y + dy$. In scenarios 1 and 2, we obtain censored observations of y^c and $(y+dy)^c$. Theorem 1 indicates that, in this case, compared with scenario 1, having additional inventory dy in scenario 2 still misses the true demand information and brings a negligible informational value (i.e., a value of only $o(dy)$).

Based on Theorem 1, we present two propositions. The first proposition compares the optimal inventory level with the myopic inventory level, which optimizes only the expected cost in the current period and equals $\arg\min_{y\geqslant 0} C_t(y, \pi_t) = M_t^{-1}((p-c)/(p-h))$. Because $S^*_t(\pi_t)$ solves $G'_t(y, \pi_t) = 0$, the following comes immediately from (9) and (10):

Proposition 2. *The myopic inventory level is a lower bound of the optimal inventory level, i.e.,* $M_t^{-1}((p-c)/(p-h)) \leqslant S^*_t(\pi_t)$.

We have established the above result for any parametric demand distribution with any prior distribution (not necessarily conjugate). Lariviere and Porteus (1999) obtained the same result assuming demand follows a newsvendor distribution. Ding et al. (2002) proved this property for a two-period model under the assumption that $f(x \mid \theta)$ is likelihood-ratio increasing in θ (also see Lu et al. 2005). Our result is stronger than that of Harpaz et al. (1982), in a similar way to that discussed by Ding et al. (2002).

Clearly, the myopic solution ignores the impact of learning from observed sales on future periods, whereas the optimal solution uses the information to improve the demand-distribution updating and thus reduce the future cost $E[V_{t+1}(\pi_{t+1})] = EG_{t+1}(S^*_{t+1}(\pi_t, y \otimes Z_t), \pi_t, y \otimes Z_t)$. In fact, under the optimal policy, the higher the inventory level is, the more the reduction of the future cost (or equivalently, the larger the informational value) will be. The second proposition quantifies the marginal informational value obtained from increasing the inventory level by dy, which follows immediately from Theorem 1 and (9).

Proposition 3. *The expected marginal informational value of increasing the inventory level by dy is*

$$-\frac{dEG_{t+1}(S^*_{t+1}(\pi_t, y\otimes Z_t), \pi_t, y\otimes Z_t)}{dy} = \left[J_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1}) - G_{t+1}(S^*_{t+1}(H^e_{t+1}), H^e_{t+1})\right] m_t(y) \geqslant 0.$$

We remark that Proposition 3 corrects the expected marginal informational value expression presented in Ding et al. (2002). In terms of our notation, their result is[1]

$$\left[G_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1}) - G_{t+1}(S^*_{t+1}(H^e_{t+1}), H^e_{t+1})\right] m_t(y). \tag{11}$$

We note that $G_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1})$ in (11) differs substantially from $J_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1})$ as defined previously. Specifically, although $G_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1})$ and $J_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1})$ imply the same decision, $S^*_{t+1}(H^c_{t+1})$, being adopted in period $t+1$, decisions starting from period $t+2$ are different: $G_{t+1}(S^*_{t+1}(\widehat{H}_{t+1}), H_{t+1})$ implies order quantities are optimally placed according to the actual information obtained by then, whereas this is not the case for $J_{t+1}(S^*_{t+1}(\widehat{H}_{t+1}), H_{t+1})$. For example, given the same observation of $S^*_{t+1}(H^c_{t+1}) \otimes z_{t+1}$ in period $t+1$, the order quantity given by $G_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1})$ is $S^*_{t+2}(H^e_{t+1}, S^*_{t+1}(H^c_{t+1}) \otimes z_{t+1})$ in period $t+2$, whereas that by $J_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1})$ is $S^*_{t+2}(H^c_{t+1}, S^*_{t+1}(H^c_{t+1}) \otimes z_{t+1})$. Because it is easy to show $G_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1}) \leqslant J_{t+1}(S^*_{t+1}(H^c_{t+1}), H^e_{t+1})$, we conclude that the expression shown in (11) underestimates the value of information gained by the optimal policy.

4. Electronic Companion

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Endnote

1. To compare the informational value with the unit shortage-penalty cost p, Ding et al. (2002) have also scaled the informational value by $1 - M_t(y)$, the probability of the inventory being out of stock.

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